Mir Mynul Ahasan

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| LinkedIn | GitHub|

EDUCATION

Bachelor of Science — *Computer Science and Engineering* - SEPTEMBER 2024
Daffodil International University, (DIU), Dhaka, Bangladesh. CGPA: 3.50/4.00
Website https://daffodilvarsity.edu.bd/

Field of study: Problem Solving, OOP, Data Structures & Algorithms, Data Communication, DBMS, Computer Network, Software Engineering, Artificial Intelligence, Data Mining, Statistics, Linear Algebra, Big Data & IoT.

Final grade: 3.50/4.00 | Level in EQF EQF level 6 | Type of credits Bangladeshi Credit System |Number of credits 148 |

SKILLS

-Programming Languages and Frameworks: C, C++ Arduino, Python, Django Rest Framework, FastAPI, Flask, JavaScript.
-Machine Learning Frameworks: PyTorch, Numpy, Pandas, Matplotlib, SciKit-Learn, HuggingFace
-Algorithm Proficiency: CNN, GNN, BERT-BiGRU, LLMs, Explainable AI, Supervised Learning (Regression, Logistic, SVM, Decision Tree, Random Forest, KNN), Image to Feature Extraction, Data Processing, and Feature engineering.
-Soft Skills: Leadership, Team Management, Event Planning, Business & Market Analysis.
-Technical skills: Postman, Swagger UI, ClickUp.

WORK EXPERIENCE

NidusLab-USA, Banani, Dhaka, BD — *AI Software Engineer* **MAY 2025 – PRESENT**

- Designed real-time communication tools for seamless teacher-parent updates, reducing workload and boosting engagement.
- Leveraged data insights to enhance student outcomes and demonstrate ROI.
- Built an all-in-one solution, consolidating 5-7 tools into one platform for cost savings.
- Developed with Python, Django, and Django REST Framework for scalable global deployment.
- Integrated JWT Authentication for secure user access and authorization.
- **Developed NidusLab RAG AI Agent** integrating PyTorch RAG and FAISS vector search for efficient information retrieval and response generation.
- Integrated **Google Gemini AI** for enhanced conversational capabilities, enabling smarter interactions.
- Employed **PyTorch + Sentence Transformers** for natural language understanding and improved conversational quality.
- Implemented **conversation memory** to maintain context and improve user experience during long-term engagements.

Diu Embedded System Research Center, Daffodil International University—*ML Researcher* **MAY 2023 – JULY 2024**

- Developed meta learning framework for medical image denoising using self-supervised learning.
- Implementing multi-class classification for chest X-ray images.
- Designing the DeepChestGNN model for graph-based chest X-ray analysis.
- Conducting image feature extraction using CNNs and handcrafted methods.
- Implemented Large Language Model (LLM) for automatic answer evaluation for the system.

PROJECT

1. SchoolConnect |

- Designed real-time communication tools for seamless updates between teachers and parents, reducing workload and boosting engagement.
- Utilized data-driven insights to provide schools with comprehensive analytics, improving student outcomes and showcasing ROI.
- Built an all-in-one solution that consolidates multiple tools into a single platform, saving schools from paying for 5-7 separate tools.
- Developed with Python, Django, and Django REST Framework, ensuring robust backend performance and scalability for global deployment.
- Integrated JWT Authentication to provide secure user authentication and authorization.

2. Multi-Persona (LLMs) AI Chatbot with Google Gemini 2.5 Flash |

- Three Unique Personas: Developed an AI chatbot with Viper (sarcastic & aggressive), Seraphina (balanced & humorous), and Elysia (seductive & engaging) conversational styles.
- Integrated **Google Gemini API** for enhanced conversational capabilities, enabling smarter interactions.
- Built a FastAPI backend to handle persona-driven prompts, streaming responses, and secure API integration.
- Designed a lightweight React frontend with persona selection, real-time messaging, and clean UI/UX.

3. NidusLab RAG AI Agent |

- **Developed NidusLab RAG AI Agent** integrating PyTorch RAG and FAISS vector search for efficient information retrieval and response generation.
- Integrated **Google Gemini AI** for enhanced conversational capabilities, enabling smarter interactions.
- Implemented **conversation memory** to maintain context and improve user experience during long-term engagements.
- Employed **PyTorch + Sentence Transformers** for natural language understanding and improved conversational quality.
- Utilized **FastAPI** for building fast, scalable APIs and **SQLite with SQLAlchemy** for efficient database management.

LICENSE AND CERTIFICATITONS AND COURSES

- **Machine Learning Specialization:** Stanford University & Deep Learning. AI (Coursera) Completed a three-course program covering supervised learning, unsupervised learning, recommender systems, and reinforcement learning. Gained hands-on experience in building and optimizing models to solve real-world problems. | **Show Credential**
- **Introduction to JavaScript:** Gained foundational knowledge in programming, including understanding data types and variables. Learned to write conditional statements and effectively use arrays. Developed coding skills by applying functions, operators, and loops. Acquired hands-on experience with object-oriented programming concepts and their practical implementation. | **Show Credential**
- **Introduction to Python Programming:** Acquired fundamental programming knowledge, including data types, variables, conditional statements, arrays, functions, operators, and loops. Gained practical experience with object-oriented programming concepts. | **Show Credential**
- **Data Structures and Algorithms:** Studied traditional data structures and their runtime complexities. Gained expertise in algorithm correctness and analysis. Conducted in-depth analysis of data structures and algorithms. Designed new algorithms and explored their applications in popular software and tools. Acquired skills in strategic debugging and testing for efficient problem-solving.

LANGUAGE

- **Bangla-Native, English-Fluent**

EXTRA CURRICULUM ACTIVITIES

Volunteer at the 4th International RoboTech Olympiad 2024

- Organizing Team – Drone Challenge: Planned and executed the Drone Challenge competition, ensuring smooth operations and participant engagement in a high-impact robotics event.

Event Organizer DIU Intra-University Robotics and Project Competition 2022

- Organized and managed a university-wide robotics and project competition, fostering innovation and technical skill development among participants. Played a key role in event planning, coordination, and execution.

DIU Embedded System Research Lab - May 2023 – July2024

- Associate Member (Researcher) - Worked on Machine Learning & Natural Language Processing with real life dataset over 2 thousand and implemented Large Language Model for automatic answer script evaluation in my thesis.

REFERENCES

- Dr. Naznin Sultana

  Associate Professor

  Department of Computer Science and Engineering.

  Daffodil International University.

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- Md. Sadekur Rahman

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